

Question 1

For the CMOS inverter with a power supply voltage of $V_{DD} = 5 \text{ V}$, determine the fall time τ_{fall} which is defined as the time elapsed between the time point at which $V_{\text{out}} = V_{90\%} = 4.5 \text{ V}$ and the time point at which $V_{\text{out}} = V_{10\%} = 0.5 \text{ V}$. Use both the average-current method and the differential equation method for calculating output load capacitance is 1 pF . The nMOS transistor parameters are given as

$$\mu_n C_{\text{ox}} = 20.0 \text{ } \mu\text{A/V}^2$$

$$\left(\frac{W}{L}\right)_n = 10$$

$$V_{T_n} = 1 \text{ V}$$

Question 2

Consider the CMOS inverter circuit with $V_{DD} = 3.3 \text{ V}$. The I-V characteristics of the nMOS transistor are specified as follows: when $V_{GS} = 3.3 \text{ V}$, the drain current reaches its saturation level $I_{\text{sat}} = 2 \text{ mA}$ for $V_{DS} \geq 2.5 \text{ V}$. Assume that the input signal applied to the gate is a step pulse that switches instantaneously from 0 V to 3.3 V . Using the data above, calculate the delay time necessary for the output to fall from its initial value of 3.3 V to 1.65 V , assuming an output load capacitance of 300 fF .

Question 3

A company in Urbana, IL called Prairie Technology has access to a CMOS fabrication process with the device parameters listed below.

$$\mu_n C_{\text{ox}} = 120 \text{ } \mu\text{A/V}^2$$

$$\mu_p C_{\text{ox}} = 60 \text{ } \mu\text{A/V}^2$$

$L = 0.6 \text{ } \mu\text{m}$ for both nMOS and pMOS devices

$$V_{T0,n} = 0.8 \text{ V},$$

$$V_{T0,p} = -1 \text{ V},$$

$$W_{\text{min}} = 1.2 \text{ } \mu\text{m}$$

Design a CMOS inverter by determining the channel widths W_n , and W_p of the nMOS and pMOS transistors, to meet the following performance specifications.

- $V_{\text{th}} = 1.5 \text{ V}$ for $V_{DD} = 3 \text{ V}$,
- Propagation delay times $\tau_{\text{pHL}} \leq 0.2 \text{ ns}$ and $\tau_{\text{pLH}} \leq 0.15 \text{ ns}$
- A falling delay of 0.35 ns for an output transition from 2 V to 0.5 V , assuming a combined output load capacitance of 300 fF and ideal step input.

Question 4

Consider a CMOS inverter, with the following device parameters:

$$\text{nMOS} \quad V_{T0,n}=0.8 \text{ V}, \quad \mu_n C_{ox}=50 \text{ } \mu\text{A/V}^2$$

$$\text{pMOS} \quad V_{T0,p}=-1.0 \text{ V}, \quad \mu_p C_{ox}=20 \text{ } \mu\text{A/V}^2$$

The power supply voltage is $V_{DD}=5 \text{ V}$. Both transistors have a channel length of $L_n=L_p=1 \text{ } \mu\text{m}$. The total output load capacitance of this circuit is $C_{out}=2 \text{ pF}$, which is independent of transistor dimensions.

- Determine the channel width of the nMOS and the pMOS transistors such that the switching threshold voltage is equal to 2.2 V and the output rise time is $\tau_{rise}=5 \text{ ns}$.
- Calculate the average propagation delay time τ_p for the circuit designed in (a).
- How do the switching threshold V_{th} and the delay times change if the power supply voltage is dropped from 5 V to 3.3 V . Provide an interpretation of the results.

Question 5

Consider a CMOS inverter with following parameters. The switching threshold is designed to be equal to 2.4 V .

$$\text{nMOS} \quad V_{T0,n}=0.8 \text{ V}, \quad \mu_n C_{ox}=50 \text{ } \mu\text{A/V}^2$$

$$\text{pMOS} \quad V_{T0,p}=-1.0 \text{ V}, \quad \mu_p C_{ox}=20 \text{ } \mu\text{A/V}^2$$

A simplified expression of the total output load capacitance is given as:

$$C_{out}=500 \text{ fF} + C_{db,n} + C_{db,p}$$

The power supply voltage is $V_{DD}=5 \text{ V}$. Both transistors have a channel length of $L_n=L_p=1 \text{ } \mu\text{m}$. Furthermore, we know that the drain-to-substrate parasitic capacitances of the nMOS and the pMOS transistors are functions of the channel width. A set of simplified capacitance expressions are given below.

$$C_{db,n}=100 \text{ fF} + 9W_n$$

$$C_{db,p}=80 \text{ fF} + 7W_p$$

where W_n and W_p are expressed in μm .

- Determine the channel width of both transistors such that the propagation delay τ_{pHL} is smaller than 0.825 ns .
- Assume now that the CMOS inverter has been designed with $(W/L)_n=6$ and $(W/L)_p=15$, and that the total output load capacitance is 250 fF . Calculate the output rise time and fall time using the average current method.